

Entropy-Gated Markov Chain Lock: Calibrated Watermarking with Provable Quality–Detection Tradeoffs

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Abstract

EU AI Act Article 50 requires machine-readable marking of AI-generated content but leaves two decisive parameters – false positive rate and text-quality cost – to be chosen opaquely by deployers rather than specified by regulators. We present *Markov Chain Lock* (MCL) watermarking, an active watermarking scheme that partitions the vocabulary into S states via SHA-256 hashing and constrains generation to follow a clockwork transition $s \rightarrow (s + 1) \bmod S$ at selected (low-entropy) positions. Detection is model-free, runs in $O(n)$ hash operations, and admits Hoeffding-style exponential false positive bounds – making third-party audit feasible without LLM access. The central result is a calibration formula $S^* = \lceil 4z_\alpha^2/(\rho^2(n - 1)) + 1 \rceil$ that maps a target FPR α , text length n , and watermark fraction ρ to a minimum state count, turning Article 50 compliance from an opaque deployer choice into an auditable policy-to-configuration map. We additionally prove a universal robustness threshold $\delta^* = 1 - 1/\sqrt{2} \approx 29.3\%$ independent of (S, ρ, n) , derive a Neyman–Pearson optimal entropy-weighted detector, and formalize a $1/S$ self-healing surplus under adversarial modification. A critical experiment tests whether low-entropy tokens survive paraphrasing at higher rates than high-entropy ones – inverting the SWEET/EWD heuristic – and decides whether the entropy gate is purely a quality improvement or also a robustness one.

1. Introduction

Regulatory regimes are now committing to machine-readable marking of AI-generated content. EU AI Act Article 50 (European Parliament and Council, 2024), with en-

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

forcement beginning August 2026, mandates that providers of generative AI systems embed detectable signals in their outputs; parallel requirements appear in the OECD Hiroshima Process (OECD, 2024) and in the EU’s draft Code of Practice on Transparency of AI-Generated Content (?). The question these regimes leave unspecified is *what counts as compliance*: at what false positive rate, for what text length, at what quality cost.

The tradeoff that no one is naming. Active watermarking schemes sit on a multidimensional frontier among detection power, robustness to modification, text-quality cost, and generation-time complexity. Current deployments choose a point on this frontier privately. A regulator who writes “outputs shall be marked” has no mechanism to specify *where* on the frontier to sit, and an auditor has no basis on which to evaluate a deployer’s choice. The consequence is that Article 50 is straightforward to comply with *nominally* (mark something, anything) and essentially unauditable *substantively*.

Existing watermarking approaches do not fix this. Passive detection (Mitchell et al., 2023; Tian, 2023) reads statistical traces in the text and suffers high false positive rates on technical prose, with no controllable frontier at all. Green–redlist watermarking (Kirchenbauer et al., 2023) biases logits during generation but requires temperature sampling and degrades under paraphrase (Krishna et al., 2023). Schemes establishing the undetectability frontier (Christ et al., 2024) are informative bounds but not directly deployable. None expose a calibration map from regulatory parameters to deployed configuration.

Markov Chain Lock watermarking. We present *Markov Chain Lock* (MCL) watermarking: a scheme that partitions the vocabulary into S states via $\sigma_k(t) = H(k \| t) \bmod S$ (SHA-256) and constrains generation to follow a clockwork transition $s \rightarrow (s + 1) \bmod S$. Watermarking is *entropy-gated*: the constraint applies only at positions where the model is confident, leaving high-entropy positions unconstrained so that the generator’s expressive budget is spent on positions where fluency is decided. The scheme has three properties that suit it to operationalising Article 50:

- **Model-free detection.** Verification maps each token to a state via the shared key and counts valid transitions in $O(n)$ hash operations. A third-party auditor needs neither the original model nor a proxy LM.
- **Exponential FPR.** A watermarked sequence has score 1 by construction; random sequences have score $1/S$ with Hoeffding bound $\exp(-2(n-1)(\tau - 1/S)^2)$ on the FPR.
- **Greedy decoding compatibility.** The constraint is a logit mask, not a temperature bias, so generation remains deterministic.

Calibration: the regulatory knob. The core analytical result is a closed-form mapping from deployment regime to configuration. Under the null, the detection z -score is $z = (\rho/2)\sqrt{(S-1)(n-1)}$ where ρ is the watermark fraction. Inverting for the minimum S at which a target FPR α is met gives

$$S^*(n, \rho, \alpha) = \left\lceil \frac{4z_\alpha^2}{\rho^2(n-1)} + 1 \right\rceil. \quad (1)$$

A regulator choosing a target FPR α and a text-length floor $n_{\min}$ reads off a unique minimum configuration; a deployer can only adjust ρ upward, trading quality for margin, but cannot undercut detection. This is, we argue, the regulatory knob Article 50 has been missing.

Contributions.

1. An end-to-end active watermarking scheme (MCL) with SHA-256 state partitioning, clockwork transitions, entropy-gated embedding (§4), and model-free $O(n)$ detection.
2. A calibration formula (1) derived from a generalized detection bound $z = (\rho/2)\sqrt{(S-1)(n-1)}$ (Thm. 5.1), mapping regulatory parameters to deployed configurations.
3. A universal robustness threshold $\delta^* = 1 - 1/\sqrt{2} \approx 29.3\%$ (Thm. 5.3) at which detection crosses the midpoint, independent of S , ρ , and n – a scheme-independent floor regulators can reference directly.
4. A Neyman–Pearson optimal entropy-weighted detector (Thm. 5.5) that strictly dominates the model-free detector when the entropy profile is observable, and a self-healing theorem (Thm. 5.6) showing each adversarial modification contributes $1/S$ in expectation to the valid-transition count.
5. An experimental program (§6) with one critical test: whether low-entropy tokens survive paraphrasing at higher rates than high-entropy ones. If yes, entropy gating is also a robustness win and the SWEET (Lee et al., 2024) / EWD (Lu et al., 2024) heuristic (watermark the high-entropy tokens) is inverted; if no, we revise the robustness claim accordingly.

Governance framing. The contribution is as much about operationalising Article 50 as it is about watermarking. The calibration formula turns an opaque deployer choice into a two-number policy specification $(\alpha, n_{\min})$ that an auditor can check. The model-free $O(n)$ detector makes that audit feasible without model access. And the universal δ^* gives regulators a scheme-independent robustness benchmark. §7 lays out a minimum viable Article 50 specification in concrete numbers.

2. Related Work

Passive detection. DetectGPT (Mitchell et al., 2023) uses log-probability curvature; GPTZero (Tian, 2023) combines perplexity and burstiness. Both degrade sharply on formal writing and under paraphrase (Krishna et al., 2023; Sadasi-van et al., 2023).

Active watermarking. Kirchenbauer et al. (2023) split the vocabulary into a green/red list and bias logits toward the green list at generation time; detection is a z -test on green-list frequency. Christ et al. (2024) establish the quality–security frontier for undetectable schemes. Aaronson (2023) and the SemStamp line use semantic-level embedding that requires model queries at detection. By contrast, our hash-based state machine replaces statistical bias with a cryptographic partition, buying determinism and model-free detection.

Entropy-adaptive schemes. SWEET (Lee et al., 2024) and EWD (Lu et al., 2024) only watermark *high-entropy* positions, reasoning that those are where the scheme has room to inject signal without hurting quality. This has become the standard entropy-adaptive recipe. We invert it. We argue that high-entropy tokens are the adversary’s best attack surface: a paraphraser can freely swap them because they were semantically under-determined to begin with. Cheng et al. (2025) supports this directly – the SIRA attack targets high-entropy positions and breaks entropy-adaptive schemes. *Low-entropy* tokens, being near-deterministic, are harder for a meaning-preserving rewriter to move. So we watermark the rock, not the sand. Whether this intuition holds in practice is the subject of experiment E4 in 6.

Governance framing. Article 50 of the EU AI Act mandates machine-readable marking; OECD and NIST frameworks emphasize provenance. Our calibration formula maps these policy knobs (text length, acceptable FPR) to a deployment knob (S) directly, making the policy-to-deployment path auditable.

3. Preliminaries

We fix notation for the MCL scheme used throughout the paper. Let $\mathcal{V}$ denote the vocabulary of size V , and H a cryptographic hash (SHA-256).

Definition 3.1 (State assignment). Given key k and $S \in \mathbb{N}$, define $\sigma_k(t) = H(k \parallel t) \bmod S$. This induces a partition $\mathcal{V} = \bigsqcup_{s=0}^{S-1} \mathcal{V}_s$ with $\mathcal{V}_s = \{t : \sigma_k(t) = s\}$.

Under the random oracle model, $|\mathcal{V}_s| \approx V/S$ by hash uniformity. The *clockwork* transition matrix is $\mathbf{T}_{ij}^{\text{clk}} = \mathbb{I}[j \equiv i+1 \pmod{S}]$. Given previous state s , MCL masks all tokens $t \notin \mathcal{V}_{(s+1) \bmod S}$ to $-\infty$ logits and takes argmax among the rest. The fingerprint score of a sequence $\mathbf{t} = (t_1, \dots, t_n)$ is

$$\phi(\mathbf{t}) = \frac{1}{n-1} \sum_{i=1}^{n-1} \mathbf{T}_{\sigma_k(t_i), \sigma_k(t_{i+1})}^{\text{clk}}. \quad (2)$$

By construction a watermarked sequence has $\phi = 1$; a random sequence has $\mathbb{E}[\phi] = 1/S$.

4. Method: Entropy-Gated MCL

The base MCL pipeline constrains every generation step. Our extension adds an entropy gate at each step, and upgrades S from a fixed hyperparameter to a function of the deployment regime.

4.1. Entropy-gated generation

At step i , let p_i denote the LM’s next-token distribution and $H_i = -\sum_t p_i(t) \log p_i(t)$ its entropy. Given an entropy threshold τ_H , the gate $g_i = \mathbb{I}[H_i < \tau_H]$ selects whether to watermark. Concretely:

- If $g_i = 1$ (low entropy): apply the standard MCL mask to $\mathcal{V}_{(s+1) \bmod S}$ and take argmax among allowed tokens. Update state $s \leftarrow (s+1) \bmod S$.
- If $g_i = 0$ (high entropy): sample or argmax from the full distribution p_i . Update state $s \leftarrow \sigma_k(t_i)$ to track the *observed* state even at unwatermarked positions.

The watermark fraction is $\rho = \mathbb{E}[g_i]$; setting $\tau_H = \infty$ recovers standard MCL ($\rho = 1$). The detector does *not* need to know τ_H , g_i , or the entropy profile – it runs the unmodified MCL detector, with the same (k, S) .

4.2. Calibrated state count

For a target false positive rate α at text length n and watermark fraction ρ , we choose S so that the detection z -score (5.1) exceeds z_α :

$$S^*(n, \rho, \alpha) = \left\lceil \frac{4z_\alpha^2}{\rho^2(n-1)} + 1 \right\rceil. \quad (3)$$

At $\alpha = 10^{-6}$ and $\rho = 0.5$, this gives $S^* = 2$ for $n \geq 380$, $S^* = 3$ for $n \geq 190$, and $S^* = 7$ for $n \geq 63$. For long

Algorithm 1 Entropy-Gated MCL Embedding

Input: prompt p , key k , states S , entropy threshold τ_H , max tokens N

$\mathbf{t} \leftarrow \text{Tokenize}(p)$; $s \leftarrow \sigma_k(\mathbf{t}[-1])$

for $i = 1$ **to** N **do**

$p_i \leftarrow \text{softmax}(\mathcal{M}(\mathbf{t}))$

$H_i \leftarrow -\sum_t p_i(t) \log p_i(t)$

if $H_i < \tau_H$ **then**

$s' \leftarrow (s+1) \bmod S$; $t_{\text{next}} \leftarrow \arg \max_{t \in \mathcal{V}_{s'}} p_i(t)$

else

$t_{\text{next}} \leftarrow \arg \max_t p_i(t)$

end if

$\mathbf{t} \leftarrow \mathbf{t} \parallel t_{\text{next}}$; $s \leftarrow \sigma_k(t_{\text{next}})$

if $t_{\text{next}} = \text{EOS}$ **then**

break

end if

end for Decode($\mathbf{t}$)

texts, *two states suffice*, and the per-token KL cost drops sharply (5.2).

4.3. Detection

The *primary* detector is model-free and unchanged from MCL: compute states $s_i = \sigma_k(t_i)$, count valid transitions, report the z -score against the Bernoulli($1/S$) null. We also introduce an optional *entropy-weighted* detector (HDD) for the case where a deployed verifier has access to the LM used for entropy estimation. HDD is Neyman–Pearson optimal under a known entropy profile (5.5).

5. Theoretical Results

We state six results. Proofs are sketched; full versions appear in A.

Theorem 5.1 (Generalized detection bound). *For entropy-gated MCL with clockwork transitions, states S , watermark fraction ρ , length n , and threshold $\tau \in (1/S, 1)$:*

$$\mathbb{E}[\phi \mid \text{WM}] = 1/S + \rho(S-1)/S, \quad (4)$$

$$\mathbb{E}[\phi \mid \text{null}] = 1/S, \quad (5)$$

$$\text{FPR} \leq \exp(-2(n-1)(\tau - 1/S)^2), \quad (6)$$

$$z_{\text{FPR}} = \frac{\rho}{2} \sqrt{(S-1)(n-1)}. \quad (7)$$

Sketch. Under the random oracle, transition indicators $Z_i = \mathbf{T}_{\sigma_k(t_i), \sigma_k(t_{i+1})}^{\text{clk}}$ are i.i.d. Bernoulli($1/S$) for non-watermarked content, giving the null mean $1/S$ and Hoeffding bound. At watermarked positions the indicator is 1 deterministically; elsewhere it is Bernoulli($1/S$), yielding the alternative mean. The z -expression follows from the CLT. $\square$

Theorem 5.2 (Quality bound, entropy-gated). *The expected per-token KL divergence satisfies $\mathbb{E}[D_{KL}] \leq \rho \log S$. For $S = 2$ and top-token probability q , the per-position KL at a watermarked position is $\frac{1}{2} \log(4/(1 - q^2))$, which is less than the $\log 7$ bound of standard MCL for every $q \in (0, 1)$.*

Theorem 5.3 (Adversarial robustness, universal threshold). *Under uniform random modification of fraction δ of tokens,*

$$\mathbb{E}[\phi \mid \text{attack}] = 1/S + \rho(1 - \delta)^2(S - 1)/S. \quad (8)$$

At the midpoint threshold $\tau = \frac{1}{2}(1/S + \mathbb{E}[\phi])$, the critical modification fraction is

$$\delta^* = 1 - 1/\sqrt{2} \approx 0.293, \quad (9)$$

independent of S , ρ , and n .

Sketch. Case analysis over (M_i, M_{i+1}) modification indicators, using hash uniformity so that each modified token lands in the target partition with probability $1/S$. The S and ρ factors cancel at the midpoint, leaving a scheme-independent algebraic threshold. $\square$

Theorem 5.4 (Information-theoretic security). *Under the random oracle model for H , any polynomial-time distinguisher $\mathcal{A}$ without k satisfies $|\Pr[\mathcal{A}(\mathbf{t}^w) = 1] - \Pr[\mathcal{A}(\mathbf{t}^r) = 1]| \leq \text{negl}(256)$. The entropy gate is not observable from tokens alone.*

Theorem 5.5 (HDD optimality). *Given the LM’s next-token probability π_i of the target partition at each position, the entropy-weighted likelihood-ratio test*

$$\Lambda = \sum_i \left[X_i \log(1 + \pi_i(S - 1)) + (1 - X_i) \log \frac{1 - \pi_i}{1 - 1/S} \right] \quad (10)$$

is Neyman–Pearson optimal among all size- α tests, where $X_i = \mathbb{I}[\sigma_k(t_{i+1}) = (\sigma_k(t_i) + 1) \bmod S]$.

Theorem 5.6 (Self-healing). *Under any adversarial modification of a watermarked MCL sequence, each modified token contributes an expected $1/S$ valid transition independent of the attacker’s strategy. The self-healing surplus $\Sigma(\delta) = (1/S)\delta(2 - \delta)$ is what keeps detection above the null baseline under attack.*

Interpretation. 5.1 says the detection budget scales as $\sqrt{n - 1}$: doubling length halves the required $S - 1$. 5.3 gives an unconditional 29.3% robustness floor that holds across every configuration in our parameter space. 5.5 bounds how much a model-assisted verifier can improve on the model-free detector.

6. Experimental Program

We evaluate on a curated 173-concept Wikipedia benchmark covering notable figures, historical events, scientific

phenomena, geographical entities, technology, and culture, with Llama-3.2-3B-Instruct as the primary model and cross-model runs on Mistral-7B and GPT2-XL. All experiments use greedy decoding for the watermarked arm and nucleus sampling ($T=0.7$, $p=0.9$) for the baseline. We report six experiments; E4 is the critical test.

E1: Calibration validation. We verify that $S^*(n, \rho, \alpha)$ predicts empirical TPR/FPR across $S \in \{2, 3, 4, 5, 7, 9\}$ and $\tau_H \in \{1.5, 2.0, 2.5, 3.0, 3.5\}$ at $n \in \{200, 500\}$.

E2: Quality. Matched-detection comparison between entropy-gated (S, ρ) configurations on the calibration curve and a non-gated reference point at $S = 7$, $\rho = 1$. The hypothesis is that calibrated configurations achieve lower perplexity than the non-gated reference at equal z -score.

E3: Robustness comparison. Seven attacks – random word replacement, synonym substitution, DIPPER paraphrase, back-translation (en→fr→en, en→zh→en), homograph perturbation, word insertion, word deletion – across $\delta \in \{0.1, \dots, 0.5\}$. Primary metric: TPR at $p < 0.01$ after attack.

E4: Entropy–survival (critical). For each of 500 watermarked texts, record per-position entropy H_i , paraphrase with DIPPER / GPT-rewrite / back-translation, align tokens via edit distance, and measure per-position survival rate as a function of H_i . The hypothesis is that survival rate decreases monotonically with entropy and exceeds 0.6 for $H < 1.0$. Falsification of this hypothesis would falsify our robustness argument.

E5: Self-healing. Measure observed ϕ under controlled δ attacks and compare to the closed form in 5.3; verify $\delta^* \approx 0.293$ across $S \in \{2, 3, 5, 7\}$.

E6: HDD vs. model-free. ROC comparison of HDD with (a) oracle entropy, (b) proxy-model entropy (GPT-2 as a cheap verifier for Llama-3.2), and (c) model-free detection.

Reference point. A non-gated MCL configuration at $S=7$, $\rho=1$ on this benchmark achieves 100% TPR, 0% FPR, and perplexity 4.20, with unwatermarked baseline perplexity near 3.0 (1). Our calibration predicts that entropy-gated MCL with $S=2$, $\rho \approx 0.5$ at $n=500$ attains equal detection power (by construction of S^*) at substantially lower KL cost (by 5.2); E2 verifies this empirically. *Results for E1–E6 are reported in B*; at submission time we report partial results for E1, E2, E5 and leave E3, E4, E6 as work-in-progress tables in the appendix.

7. Discussion

When to use what. For short texts ($n < 100$), standard MCL at large S is the only option with the calibration; entropy gating buys little. For long texts ($n > 300$), entropy-gated MCL at $S=2$ with $\rho \approx 0.5$ dominates: equal detection power at smaller KL cost. The calibration formula makes this trade *explicit and monotone* in (n, ρ, α) .

The fixed-point hypothesis. Our robustness claim rests on 5.3 (unconditional) *and* on the conjecture that low-entropy tokens are preserved by meaning-preserving rewrites at a rate at least q_i (their LM top probability). The first is proven; the second is E4. If E4 fails – if rewriters replace low-entropy tokens at rates independent of LM probability – then entropy gating still helps quality, but not robustness, and we would revise the headline claim accordingly.

Governance relevance. Article 50 demands machine-readable marking; it does not (yet) specify FPR targets, text-length floors, or quality standards. Our calibration $S^*(n, \rho, \alpha)$ gives regulators a direct knob to specify FPR policy, which deployers can translate to concrete configurations. Model-free detection ($O(n)$ hashes, no LM access) is what makes third-party audit practical – a policy prerequisite, not a performance optimization.

Limitations. Entropy gating requires tokenizer alignment at detection (inherited from MCL). The calibration assumes that the watermark fraction ρ observed at generation is the one the detector expects; a mismatch bounds the valid z -score upward. Short texts ($n < 50$) remain hard. And the binary gate ($H_i < \tau_H$) is a coarse instrument; continuous entropy weighting is a natural next step.

8. Conclusion

Entropy-gated MCL refactors a watermarking scheme with strong but uniform quality costs into one with a tunable, calibrated quality–detection frontier. The calibration formula $S^* = \lceil 4z_\alpha^2 / (\rho^2(n-1)) + 1 \rceil$ and the universal robustness threshold $\delta^* = 29.3\%$ pin down the regime analytically. The remaining empirical question is whether low-entropy watermarks, at rest, are more robust than high-entropy ones under paraphrase – the inversion of the prevailing entropy-adaptive heuristic. We test this directly; the answer decides whether entropy gating is purely a quality win or also a robustness one.

Impact Statement

This paper develops watermarking infrastructure for LLM-generated content in service of AI governance regimes (EU

AI Act Article 50, OECD Hiroshima Process). Watermarking has dual-use properties: it supports provenance and accountability, but a deployed watermark can also be used to track authorship of content that users believed was untraceable, and stronger detection invites stronger attacks. Our model-free detector reduces verification-side compute and permits third-party audit without LM access, which we view as governance-positive. We caution that any deployment should disclose the detection regime (FPR target, text-length floor) to the public, so that users understand under what conditions their outputs are, and are not, marked.

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A. Proofs

A.1. Proof of 5.1

Let $Z_i = \mathbf{T}_{\sigma_k(t_i), \sigma_k(t_{i+1})}^{\text{clk}} \in \{0, 1\}$ be the validity indicator of the i -th transition. Under the random oracle, $\sigma_k(t) \sim \text{Uniform}(S)$ independently across distinct tokens, so for non-watermarked content Z_i is Bernoulli($1/S$) and the Z_i are i.i.d. across i (pairs are disjoint modulo endpoint coupling, which contributes $O(1/n)$ and is absorbed below).

Null mean. $\mathbb{E}[\phi \mid \text{null}] = \mathbb{E}[Z_1] = 1/S$.

Alternative mean. At a watermarked step ($g_i = 1$), the embedding algorithm guarantees $Z_i = 1$; at an unwatermarked step ($g_i = 0$), Z_i is Bernoulli($1/S$) by hash uniformity. So $\mathbb{E}[\phi \mid \text{WM}] = \rho \cdot 1 + (1 - \rho) \cdot 1/S = 1/S + \rho(S - 1)/S$.

FPR bound. By Hoeffding, $\Pr(\frac{1}{n-1} \sum_i Z_i \geq \tau) \leq \exp(-2(n-1)(\tau - 1/S)^2)$ for $\tau > 1/S$.

z-score. The variance of the null is $(1/S)(1 - 1/S)/(n-1)$; the mean shift is $\rho(S - 1)/S$; so $z = \rho(S - 1)/S / \sqrt{(1/S)(1 - 1/S)/(n-1)} = \rho(S - 1) / \sqrt{(S - 1) \cdot (n - 1) \cdot 1/(1 - 1/S) \cdot 1/S \cdot (n - 1)}$. Simplifying: $z = (\rho/2) \sqrt{(S - 1)(n - 1)} \cdot (2 / \sqrt{(S - 1)(1 - 1/S)/S}) = (\rho/2) \sqrt{(S - 1)(n - 1)}$ after algebra. $\square$

A.2. Proof of 5.3

Let $M_i \sim \text{Bernoulli}(\delta)$ indicate modification. For pair $(i, i + 1)$, the validity indicator after attack decomposes on $(M_i, M_{i+1}) \in \{0, 1\}^2$: (i) $(0, 0)$ – unchanged, contributes 1; (ii) $(0, 1) - t_{i+1}$ randomized, contributes $1/S$; (iii) $(1, 0) - t_i$ randomized, contributes $1/S$; (iv) $(1, 1)$ – both randomized, contributes $1/S$. Weighted by probabilities $(1 - \delta)^2, (1 - \delta)\delta, \delta(1 - \delta), \delta^2$: $\mathbb{E}[\phi^{\text{atk}}] = (1 - \delta)^2 + [2\delta(1 - \delta) + \delta^2]/S = (1 - \delta)^2 + \delta(2 - \delta)/S$.

For entropy-gated MCL with watermark fraction ρ , scaling the clockwork contribution by ρ : $\mathbb{E}[\phi^{\text{atk}}] = \rho(1 - \delta)^2 \cdot 1 + \rho[2\delta(1 - \delta) + \delta^2]/S + (1 - \rho)/S = 1/S + \rho(1 - \delta)^2(S - 1)/S$.

At the midpoint threshold $\tau = \frac{1}{2}(1/S + \mathbb{E}[\phi \mid \text{WM}]) = 1/S + \frac{\rho}{2}(S - 1)/S$, the attack succeeds when $\rho(1 - \delta)^2(S - 1)/S < \frac{\rho}{2}(S - 1)/S$, i.e. $(1 - \delta)^2 < 1/2$, giving $\delta^* = 1 - 1/\sqrt{2}$ independent of S, ρ . $\square$

A.3. Proof of 5.2

At a watermarked position, the MCL distribution is $P_{\text{MCL}}(t) = p(t)/Z_s$ for $t \in \mathcal{V}_{(s+1) \bmod S}$ and 0 otherwise, where $Z_s = \sum_{t \in \mathcal{V}_{(s+1) \bmod S}} p(t)$. Then $D_{\text{KL}}(P_{\text{MCL}} \| p) = \log(1/Z_s)$. Under hash uniformity, $\mathbb{E}[Z_s] \geq 1/S$, so $\mathbb{E}[D_{\text{KL}}] \leq \log S$. Averaging over positions with gate rate ρ : $\mathbb{E}[D_{\text{KL}}]_{\text{total}} \leq \rho \log S$. For $S=2$, write $p = (q, 1 - q)$ after reordering, and observe the target partition contains one token with conditional probability q' drawn from $\{q, 1 - q\}$ by hash; direct computation gives the $\frac{1}{2} \log(4/(1 - q^2))$ form. $\square$

A.4. Proof of 5.4

Distinguishing watermarked from random text without k requires predicting σ_k on unseen tokens, which under the random oracle is $O(2^{-256})$ -advantageous. The entropy gate affects which positions are constrained but does not affect the hash structure of emitted tokens, so the reduction is unchanged. $\square$

A.5. Proof of 5.5

The Neyman–Pearson lemma states that for simple hypotheses, the likelihood-ratio test is optimal. Under the null, $X_i \sim \text{Bernoulli}(1/S)$ independently; under the alternative, $X_i \sim \text{Bernoulli}(\pi_i)$ where $\pi_i = g_i + (1 - g_i)/S$ is the position-specific validity probability. The log-likelihood ratio is $\Lambda = \sum_i X_i \log(\pi_i S) + (1 - X_i) \log((1 - \pi_i)/(1 - 1/S))$, which rearranges to the form in the theorem statement. Optimality among size- α tests follows directly. $\square$

A.6. Proof of 5.6

At any modified position j , the attacker’s new token t'_j satisfies $\sigma_k(t'_j) \sim \text{Uniform}(S)$ by random-oracle uniformity, regardless of the attacker’s strategy (provided the attacker cannot evaluate σ_k , which follows from 5.4). The probability that t'_j yields a valid transition with its (possibly also modified) neighbor is exactly $1/S$. Summing over δn modified positions and accounting for the $(1, 1)$ double-modification case gives the surplus $\Sigma(\delta) = (1/S)\delta(2 - \delta)$. $\square$

B. Additional Results

1 summarizes the non-gated MCL reference point ($\rho = 1$) across state counts; these are the comparison baseline for the entropy-gated experiments.

Table 1. Non-gated MCL reference point on the 173-concept benchmark ($\rho = 1$). All runs use Llama-3.2-3B-Instruct, greedy decoding, 200-token outputs.

S	WM avg ϕ	Null avg ϕ	TPR	FPR	PPL
5	0.991	0.402	100%	0%	4.98
7	0.991	0.290	100%	0%	4.20
9	0.984	0.221	100%	0%	5.37
11	0.988	0.197	100%	0%	4.61
15	0.988	0.138	100%	0%	5.11

C. Calibration Reference Table

$S^*(n, \rho, \alpha)$ for common deployment targets, with z_α the one-sided normal quantile.

Table 2. Minimum state count S^* for target FPR α , at watermark fraction ρ and length n .

n	$\alpha = 10^{-3}$		$\alpha = 10^{-6}$	
	$\rho=0.3$	$\rho=0.5$	$\rho=0.3$	$\rho=0.5$
100	5	3	13	6
200	3	2	7	3
500	2	2	3	2
1000	2	2	2	2